VAAL UNIVERSITY OF TECHNOLOGY

Diploma in Information Technology

BUSINESS ANALYSIS 3 MODULE 2

AIBUY3A | PROJECT REPORT

StockSense AI

Retail sales forecasting and stock replenishment



Prepared by

VUT AI Solutions

Theme: An AI Solution for Industries

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Python application | Web dashboard | Terminal assistant

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Project overview

StockSense AI is a local Python application for sales forecasting and demonstration stock planning. Sunrise Mini Market, our fictional retailer, uses anonymous store 1 and items 1 to 5 from Kaggle. We compared a weekly benchmark, random forest and a small LSTM using chronological data splits. Random forest achieved the lowest daily validation error and remains selected after final testing. Users can choose a web dashboard or terminal chat. Both provide seven dated estimates, a transparent stock calculation and a five-intent text assistant. This report records the implemented scope and measured evidence. Real cost savings and stockout reductions require a separate business trial. The official Grammarly results are supplied as a separate PDF.

Declaration and group members

We declare that the submitted project is our own work and that we have referenced the work and ideas of others where used. Each member must review and sign this declaration before submission. The signature spaces below are intentionally blank.

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1 Background and theme relevance

StockSense AI uses historical sales to help Sunrise Mini Market, a fictional retailer, plan stock purchases. Kaggle's Store Item Demand Forecasting dataset supplies anonymous sales records [1]. The Python prototype covers store 1 and items 1 to 5, predicts seven daily quantities and calculates order suggestions using entered stock and delivery assumptions. It fits the theme An AI Solution for Industries by applying machine learning to a retail decision. The intended benefit is a consistent estimate before ordering. A local dashboard and text assistant make the estimates and assumptions accessible. The historical demonstration does not establish business performance in a South African shop.

2 Problem definition

The problem

In our fictional retail case, the manager decides how much stock to order by checking current stock and recent sales. The process does not use a consistent forecast for the period ahead. If the manager underestimates sales, products may run out before the next delivery. If the manager overestimates sales, money is tied up in stock that sells slowly. The manager needs a simple way to estimate upcoming sales and compare those estimates with available stock before placing an order. This case study assumption defines the business problem. The selected dataset provides historical sales for testing forecasts, but does not establish that a particular South African retailer experiences this problem.

The benefit of solving it

StockSense AI produces seven-day sales estimates and demonstration order suggestions. This supports the retail industry by applying AI to stock planning. The intended benefits are more consistent ordering and clearer information for the manager. Forecast accuracy was tested on historical data. Reductions in stockouts, waste or costs would require a separate inventory simulation or business trial before they could be claimed as achieved results.

For the community, more reliable stock could improve residents' access to goods and reduce repeat shopping trips. Supporting local small retailers could also benefit economic activity in the municipality. These are intended benefits, not measured outcomes.

3 Business analysis

3.1 Business background and current process

The case represents Sunrise Mini Market, a fictional small retailer that records sales each day and orders goods from a supplier. Its demonstration sales are mapped to anonymous dataset store 1. Products 1 to 5 in the app correspond to dataset items 1 to 5; their actual names remain unknown. We do not assign an actual location or product names because the dataset does not provide these details.

The assumed current process is to review recent sales, check available stock and choose an order quantity manually. In the proposed process, the manager opens the application, selects an item and checks its sales history and seven-day forecast. The manager enters stock and delivery information, then reviews the suggested quantity and its calculation. Ordering remains a business decision made by the manager.

3.2 Main objective

Develop a small Python application that forecasts seven days of sales for five items and uses those estimates to support understandable stock decisions. Compare the learned models with a weekly benchmark and retain a simple local demonstration that every member can explain.

3.3 Business objectives and success criteria

Objective	Acceptance criterion	Evidence and status
Consistent estimate	Seven dated values for all five items	Latest full suite passed on the office PC on 16 October 2026: 79 tests, including web, terminal and product-specific chatbot checks.
Lower daily forecast error	Beat weekly benchmark on identical dates	Random forest final MAE 4.812 versus 6.273 units, about 23.3% lower.
Transparent stock suggestion	At least ten arithmetic cases and invalid inputs	Automated calculation cases and dashboard checks passed.
Practical local response	Saved forecasts displayed within ten seconds	AppTest initial render 7.29 seconds; warm selections 0.20-0.23 seconds. Browser latency and human timing remain separate.
Useful text requests	At least 20 of 25 supported intents recognised	Saved fresh evaluation: 24/25, plus 8/8 unrelated requests rejected.

Lower waste, fewer stockouts and lower costs remain business objectives. The sales dataset has no inventory balances, delivery records or prices, so this prototype cannot establish those benefits.

3.4 Requirements

The system must load the sales CSV, check its required columns and organise records by store, item and date. It must let the manager choose from items 1 to 5 at store 1, show recent sales and produce a seven-day forecast. Stock inputs must lead to a visible order suggestion and explanation. The manager can use the web dashboard or terminal chat. Both interfaces must retrieve the same forecasts and apply the same stock calculation rule, with separate session inputs.

The application must reject negative quantities, unknown item IDs and unsupported delivery horizons. It must explain when insufficient history prevents a forecast. Saved models and demonstration files should allow it to run without a paid cloud service. The forecast cut-off date must remain visible so that a historical demonstration is not mistaken for a live forecast.

All sales, stock and order quantities are expressed in units. We use Python for the application and GitHub Projects to assign and track work. Each report section should have a writer and another member who reviews it.

3.5 Constraints

Our scope is one store, five items and a seven-day forecast. We will not connect the prototype to a till system or place supplier orders. The data covers 2013 to 2017, so the demonstration is historical. Kaggle's description notes that there are no holiday effects or store closures [1]. We will not claim that the data measures South African holidays, pay cycles or load shedding.

Stock levels, outstanding orders, supplier lead times and the buffer stock quantity are demonstration inputs. Prices and purchase costs are absent, which prevents a financial savings calculation. The available laptop and submission deadline also limit the size of the model comparison.

3.6 Risks and responses

Risk	Response
Future sales enter the training features	Build each sample from information available at its forecast date. Keep all seven target dates inside the assigned training period.
A complex model performs poorly	Compare every candidate with the same simple baseline and retain the more reliable method for the demonstration.
A stock recommendation is treated as a guarantee	Display its inputs and assumptions. Flag possible shortages before delivery and explain that uncertain sales can differ from the forecast.
The demonstration fails on another laptop	Save dependencies and model files, rehearse on another machine and keep a backup recording.
Report sections disagree	Use the same scope and data splits throughout and review the complete report together.

3.7 Tools and techniques

Python and pandas validate and prepare the sales records. scikit-learn supplies random forest regression and text classification. Keras with a local PyTorch backend supplies the small LSTM comparison. Streamlit combines the history, forecast, stock form and assistant on one page. A terminal interface provides the same planning and text functions without a browser. Both interfaces use the shared forecasting loader and saved models. Scripts train models separately. No paid online language service is required.

The weekly benchmark is easy to explain. Random forest uses recent sales and known calendar information. LSTM provides a small sequence experiment and a measured deep-learning comparison. The two interfaces share this logic and keep the scope explainable. Git stores source and documentation; raw data, the environment and trained models remain local. GitHub Projects tracks assigned work through Todo, In Progress and Done columns [12]. Member discussions record poster choices and meeting coordination [9, 10].

4 Machine learning approach

4.1 Prediction task

This is supervised regression: observed daily sales are the targets, and each model predicts seven quantities. Sales can understate demand when inventory is unavailable. The dataset has no stockout information, so we forecast recorded sales rather than claim unconstrained demand.

Random forest inputs are 28 sales lags, 7-, 14- and 28-day means, and sine/cosine encodings of each target date's weekday and annual position. Dates are known at the cut-off; target sales are excluded from inputs. Every fitting window keeps all seven target days inside the training boundary.

4.2 Compared methods

The weekly seasonal naive benchmark repeats the last observed week's sales on matching weekdays [3]. One random forest per item predicts seven outputs with 100 trees, depth 10, minimum leaf size 3, seed 42 and one worker [4]. These fixed settings are the implemented comparison; no tree-depth search was performed. One small LSTM per item uses 28 chronological sales inputs and seven outputs, without calendar inputs.

The predefined criterion is overall daily 2016 validation MAE, with simpler methods retained on ties. Random forest won this criterion. Selection-lock.json froze the method, dataset checksum, settings and epoch counts before 2017 scoring. The LSTM's stronger weekly-total result is reported rather than changing the criterion afterwards. Linear regression was considered earlier but was not part of the completed measured comparison.

5 Data

5.1 Source and verified structure

The supplied Kaggle archive contains train.csv, test.csv and sample_submission.csv. The office inspection on 9 October 2026 was independently reproduced on this personal PC on 10 October 2026. It contains 913,000 rows and four columns, covering 1 January 2013 to 31 December 2017. There are 10 stores and 50 items, giving 500 combinations of stores and items. Each series has 1,826 daily records.

Field	Data type	Actual example	Meaning and use
date	Temporal: calendar date	2013-01-01	Day of the record; sorts history and supplies calendar features.
store	Categorical ID, stored as an integer	1	Anonymous store identity; selects store 1.
item	Categorical ID, stored as an integer	1	Anonymous product identity; selects items 1 to 5.
sales	Numerical: discrete whole-number count	13	Units sold that day; historical input and prediction target.

Store and item numbers identify categories; a larger ID does not imply a larger store or better product. Dates have daily resolution, without a recorded time of day. These semantic data types explain the fields' meanings, while their storage formats support validation and filtering.

The chosen subset contains 9,130 records. The five items have different average sales levels, which lets us check whether a method works consistently across items rather than showing only one favourable example.

5.2 Actual sample records

These are the first five source records for store 1 and item 1.

date	store	item	sales
2013-01-01	1	1	13
2013-01-02	1	1	11
2013-01-03	1	1	14
2013-01-04	1	1	13
2013-01-05	1	1	10

5.3 Data quality checks and preparation

We found no missing cells, duplicate rows, repeated date-store-item keys or negative sales. Quantities range from 0 to 231. The complete source has one zero-sales record. Every combination of a store and an item has a complete daily history over the same date range.

Preparation preserves train.csv, selects the five series and sorts by date. Checks reject missing columns, invalid dates, duplicate daily keys, missing days and negative or fractional quantities. Zero sales remains a valid value. No missing day is silently converted to zero. Each item has 1,061 initial fitting windows and 1,427 final refit windows.

Raw and prepared SHA-256 checksums match the office record. Both SHA-256 values were checked again on 12 October 2026 and matched the preserved local audit. The raw and prepared files remain outside Git. Checksums identify the files used without placing raw data in Git.

5.4 Other data used by the application

Stock data is entered manually for the demonstration. For example, item 1 has 20 units available and 5 units already ordered. The assumed supplier lead time is 3 days, the review interval is 1 day, and the buffer stock is 10 units. These values are assumptions, not Kaggle observations.

Stock, outstanding orders and buffer quantities are numerical whole-number unit counts. Lead time and review interval are numerical whole-number day counts. The item reference remains a categorical identifier.

The assistant's training corpus contains short labelled English examples. Independent member questions and review remain pending. For example, "How much will item 2 sell next week?" is unstructured text with the categorical intent label forecast. TF-IDF converts that text to derived numerical features. The resulting answer retrieves the relevant forecast rather than invent a number.

Output records contain a categorical item ID, temporal forecast date, numerical predicted quantity and categorical model label. Predictions can contain fractions, unlike the original whole-number sales counts. Recommendation records include the stock inputs and calculated whole-number order quantity. Charts visually summarise the numerical sales and temporal dates; they are presentation outputs rather than image inputs to a model. We document these derived outputs separately from original observations.

The source contains anonymous IDs rather than customer details. We will use those IDs for this project and follow the dataset's competition terms. The raw CSV will remain separate from report drafts and application source code.

6 Model evaluation and time series analysis

6.1 Chronological evaluation

Period	Raw records for five items	Role
2013-2015	5,475	Initial fitting
2016	1,830	Validation and frozen selection
2017	1,825	Final evaluation

Each evaluation uses 52 complete weeks per item: 364 daily predictions per item, 1,820 overall and 260 item-week totals. Validation scores 1 January to 29 December 2016 and excludes 30-31 December. Final testing scores 1 January to 30 December 2017 and excludes 31 December. All methods use the same dates.

Observed history advances between forecast weeks, while learned weights remain fixed. Each forecast sees sales only before its cut-off. Final models were refitted using data through 31 December 2016, with fixed validation-selected LSTM epochs and no early stopping on 2017. Save/reload checks ensure the scored models match persisted files. The separate Kaggle 2018 test.csv has no sales column and cannot supply local accuracy scores.

6.2 Error measures

Daily MAE averages absolute predicted-minus-observed errors in units [5]. RMSE weights larger errors more strongly. Weekly-total MAE compares each predicted and observed seven-day total. Predictions are clipped at zero and scored without rounding. Percentage error reduction equals 100 times (benchmark MAE minus selected MAE) divided by benchmark MAE. It is a relative reduction in error, not a percentage of accuracy. Stock order rounding happens only after the calculation.

6.3 Observed time series patterns

The following charts use store 1 and items 1 to 5 from 2013 through 2016. We have left 2017 out of the descriptive charts used to design the models.

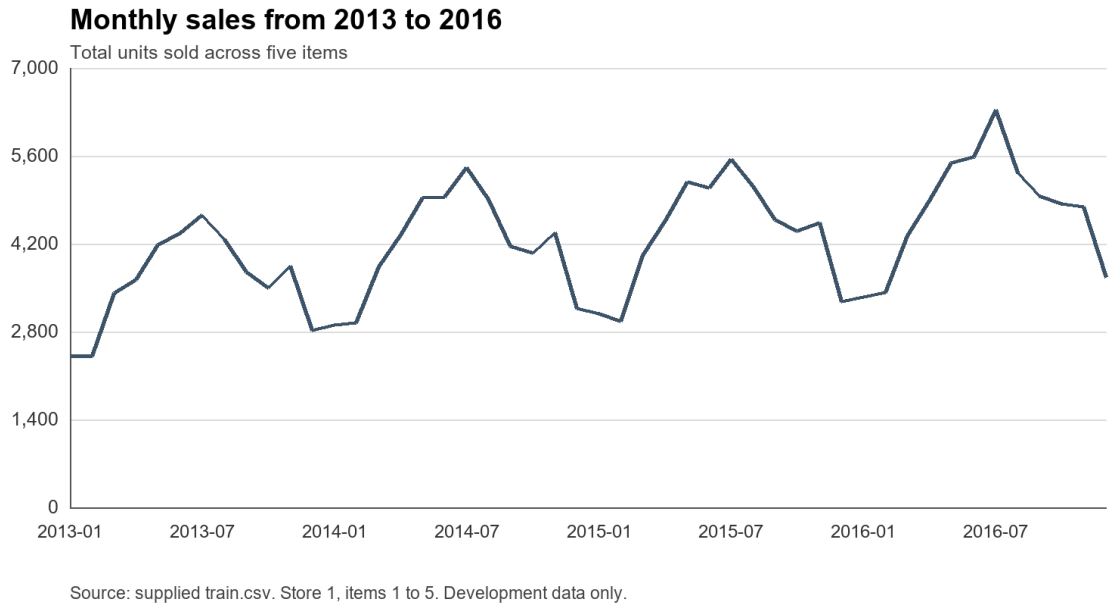


Figure 1. Monthly total units for store 1 and items 1 to 5, using development data only.

Total annual sales for these items rise from 43,208 units in 2013 to 56,943 in 2016. The monthly chart also shows repeated rises and falls across the years. We use calendar features and recent sales levels to represent these patterns. The chart describes the data; it does not establish the cause of the changes.

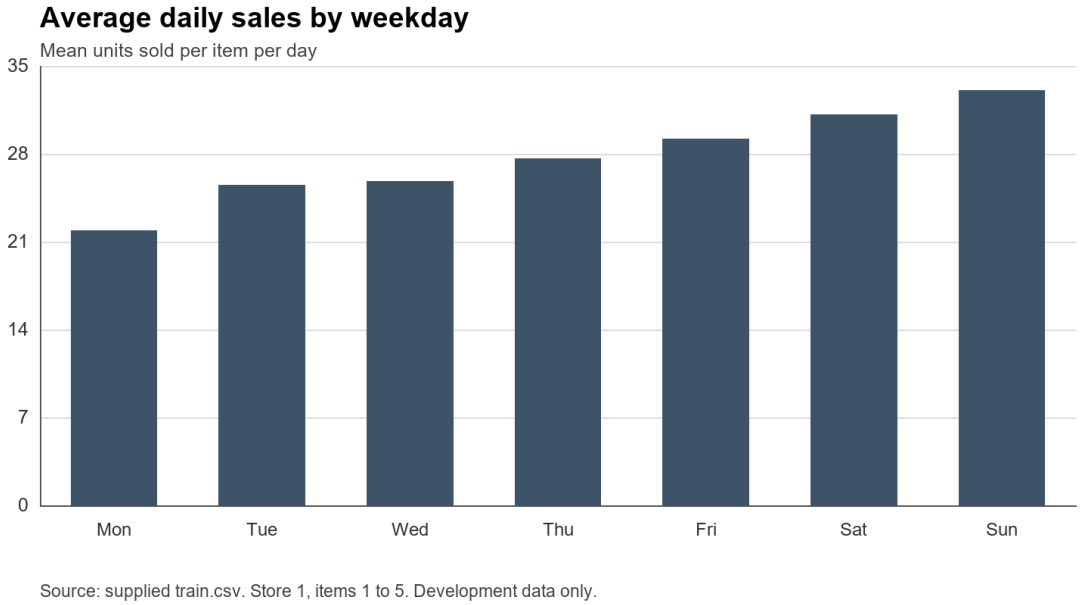


Figure 2. Mean units sold per item per day, grouped by weekday, for 2013 through 2016.

The mean is 21.88 units on Mondays and 33.01 on Sundays for this subset. This supports using a weekly seasonal baseline and weekday features. The figures combine five items, so we also report item-level errors rather than assume that each item follows the same pattern.

6.4 Measured model comparison

2016 validation on this PC:

Method	Daily MAE	Daily RMSE	Weekly-total MAE
Weekly benchmark	6.320	8.217	18.758
Random forest	4.911	6.596	19.797
Small LSTM	5.439	7.227	17.280

2017 final test on this PC:

Method	Daily MAE	Daily RMSE	Weekly-total MAE
Weekly benchmark	6.273	8.218	18.427
Random forest	4.812	6.387	17.973
Small LSTM	5.411	7.261	17.343

The personal-PC metrics match the documented office results to three decimal places. Random forest has the lowest daily error for all five items and approximately 23.3% less overall final daily MAE than the benchmark. LSTM has the lowest overall weekly-total error in both periods. Random forest's weekly-total validation error is worse than the benchmark; its final weekly error is slightly better overall but worse for items 3 and 4. Daily selection therefore does not demonstrate optimal stock orders. The methods also receive different inputs, so this limited configuration comparison cannot establish general algorithm superiority.

The full per-item results are preserved in local artifacts/forecasters/final-test.json and the historical evaluation document in Git at commit 8be24ad. Models forecast 1-7 January 2018 in the dashboard using history through 31 December 2017. Actual outcomes are absent, and neither historical table measures those estimates' future accuracy.

7 Solution techniques

7.1 Performance and fair comparison

We used recent means and known calendar encodings for random forest, chronological sales sequences for LSTM, and validation loss to choose LSTM stopping epochs. The comparison kept one declared forest configuration and one small network architecture. Test-year data did not choose features, settings, scaling or model selection. Completed evaluation files are protected against accidental retraining.

Future improvements would require a new prospectively labelled experiment and new untouched evaluation data. Inventory simulation could compare order decisions using delivery, cost and stockout information. The current 2017 results

must not become a tuning set. Human review can improve explanations without changing the frozen models.

7.2 Stock recommendation rule

The demonstration uses a simple periodic review rule. Its inputs are supplier lead time L, review interval R, stock available S, outstanding units O due within the coverage period and a fixed buffer B. We limit L plus R to seven days. Outstanding units and an order placed today arrive after the entered lead time. Lead time 2 means before forecast day 3; lead time 0 means before day 1. Outstanding units do not cover sales before that arrival.

The target stock is the sum of predicted sales over the next L plus R days, plus B. Inventory position is S plus O. The suggested order is the positive difference between target stock and inventory position, rounded upwards. If the inventory position exceeds the target, the suggested order is zero. This is a transparent planning rule rather than an optimised inventory policy.

For an illustrative forecast of 10 units per day, a three-day lead time and one-day review interval give expected sales of 40 units over the coverage period. A buffer of 10 gives a target stock of 50. If stock available is 20 and outstanding orders total 5, the order suggestion is 25 units. These values demonstrate the arithmetic and are not model results.

The interface flags a possible shortage before delivery when forecast sales before delivery exceed currently available stock. An order placed today cannot automatically resolve a shortage before the supplier arrives. We state the simplified arrival assumptions with the recommendation.

7.3 Difficult cases

An unknown item or one with less than 28 days of history triggers an explanation instead of an unsupported model forecast. Invalid stock quantities are rejected. A product with sparse sales may need a different baseline, although the selected subset will first be assessed through the common comparison. Sales without inventory information cannot reveal lost demand during stockouts, and this remains a limitation.

8 Natural language processing

The English assistant recognises forecast, reorder, history, performance and help. Processing lowercases text, normalises numbered item references and removes English stop words. TF-IDF word unigrams and bigrams supply numeric text features [6]. Logistic regression uses C=5, seed 42 and at most 1,000 iterations. A separate item check rejects unsupported IDs. For another supported item, the assistant retrieves its forecast or history; stock questions ask the user to select that product and enter its stock.

The corpus has 60 training questions and 15 validation questions. Threshold 0.35 achieved 15/15 correct validation decisions, compared with 14/15 at 0.45 and 12/15 at 0.55. The recorded classifier evaluation recognised 24/25 supported requests and rejected 8/8 unrelated requests. The missed historical-demand-summary request became forecast rather than history. The saved classifier and threshold remain unchanged. The earlier 25+8 question sets became development evidence after processing changes and remain stored separately. These small curated results measure the classifier, not every conversational feature. Additional retrieval and follow-up rules use separate regression checks. Voice and multilingual support remain outside scope.

9 Deep learning

Each item's model receives 28 chronological daily sales values, one 16-unit LSTM layer and a seven-output dense layer [7]. It has 1,271 trainable parameters and no calendar features. Train-only mean and standard deviation scale inputs and targets. Predictions return to sales units before clipping and scoring. Constant series use scale 1.

Training uses Adam at 0.001, MSE, batch size 32, no shuffling, a maximum of 50 epochs and seed 42 plus item ID. PyTorch uses CPU, one thread and deterministic algorithms. Early stopping monitors the 2016 validation loss with patience 5 and restores the best weights [8]. The frozen selected epochs for items 1-5 are 39, 50, 12, 50 and 21. Items 2 and 4 reached the declared cap, which was not extended. Final refitting recomputes scaling through 2016 and uses these fixed epochs without 2017 callbacks.

The network's overall daily final MAE is 5.411 units and weekly-total MAE is 17.343 units. Its daily error is higher than the random forest error, but its weekly-total error is lower overall. The retained experiment supports an honest deep-learning comparison.

10 Chatbot design

The assistant combines intent recognition with template-based retrieval of dashboard objects. Forecast answers use the selected item's real estimates, reorder answers use a valid submitted stock plan, history answers use observed sales, and performance answers distinguish 2016 validation from 2017 testing. No online language model invents quantities.

The improved assistant compares forecast or observed unit sales for named products or all five products, using matching date ranges. It can show seven dated forecast values, identify the highest or lowest forecast day, retrieve a weekday or date within the forecast window, and summarise a requested recent-history period. Comparisons identify tied totals and distinguish predicted demand from order quantities. Requests for live dates or longer forecast horizons explain the historical seven-day scope.

A small conversation context remembers the last product and supported topic. After a forecast for Product 2, "and Product 3?" retrieves Product 3's own estimates; "show the daily breakdown" continues with that product. A follow-up after a comparison requests one product when needed. Changing the selector or clearing the chat resets the context. Submitting a stock calculation focuses the conversation on the selected product. Order explanations show forecast demand, buffer, target stock, inventory position and rounding. Short definitions explain planning terms and model-error measures. The dataset has no prices or costs, so the assistant cannot calculate profits or real savings.

Missing stock inputs produce a request to calculate an order first. Low-confidence or unrelated questions show supported options. Explicit references to another supported item retrieve its forecast or history. Stock questions for another item request a Product selector change and fresh stock inputs. Changing items or submitting invalid coverage clears stale stock results; asking a question keeps a valid order visible. Integration tests compare answers with the actual displayed totals and check these state rules. Independent group phrasing tests remain pending.

Terminal chat uses the same trained intent classifier, forecast retrieval and order rule. Product selection uses the /product command. The /stock command prompts for the five planning inputs. /clear resets follow-up focus while keeping a valid plan for the selected product; changing products clears that plan. Terminal answers use tables and command guidance. Web answers use dashboard guidance. Each

interface keeps its own session state, so submitting an order scenario in one does not submit it in the other.

11 Practical solution and demonstration

The Streamlit page shows recent history, seven dated estimates, stock planning, and Ask StockSense. Separate expanders report validation and final testing. Launch it with python -m streamlit run app.py using the project's Python environment, then open the Local URL printed in the terminal. Terminal chat starts with python -m stocksense.terminal and requires no web server. Both load the same local data and saved models; training occurs outside the interfaces. README.md provides the complete Windows setup and launch commands.

On 10 October 2026, the personal PC reproduced the setup using Python 3.14.6. NumPy 2.3.5, pandas 3.0.1, scikit-learn 1.9.1, Streamlit 1.65.0, joblib 1.6.0, Keras 3.15.1 and PyTorch 2.14.1 match the documented package pins. The office reference used Python 3.12.14. The personal PC passed all 45 tests available at that time, with no skips. After terminal chat was added, all 79 tests in the office-PC suite passed on 13 October 2026, with no skips [11]. The full suite passed again on 16 October 2026: 79 tests in 71.22 seconds, with no skips. Checks cover web and terminal responses, product comparisons, dated forecasts, follow-up context, product-specific stock inputs, invalid values and saved-model loading. Frozen forecasting and classifier files were preserved.

The terminal demonstration reproduced Product 2's seven-day total of 310.6 units and Product 1's default 38-unit order [11]. It also checked daily tables, product switching and order explanations. These are technical checks, not independent human usability results.

The second-machine technical rehearsal checked all five items, seven forecast dates, a valid order, invalid coverage, shortage messaging, an assistant answer matching the screen, and both evaluation tables. AppTest initial render took 7.29 seconds; warm item changes took 0.20-0.23 seconds in one observed run. These timings exclude browser network latency and are not human usability evidence. Detailed logs remain on the personal PC in artifacts/rehearsal/personal-pc.json and artifacts/rehearsal/tests.txt.

Item 1's displayed forecast totals 114.9 units. With default stock 50, outstanding 0, lead time 2, review interval 3 and buffer 10, the first five predictions total 77.52 units. Target stock is 87.52 units, and the suggestion is 38 units. This is demonstration arithmetic, not an observed purchase or saving.

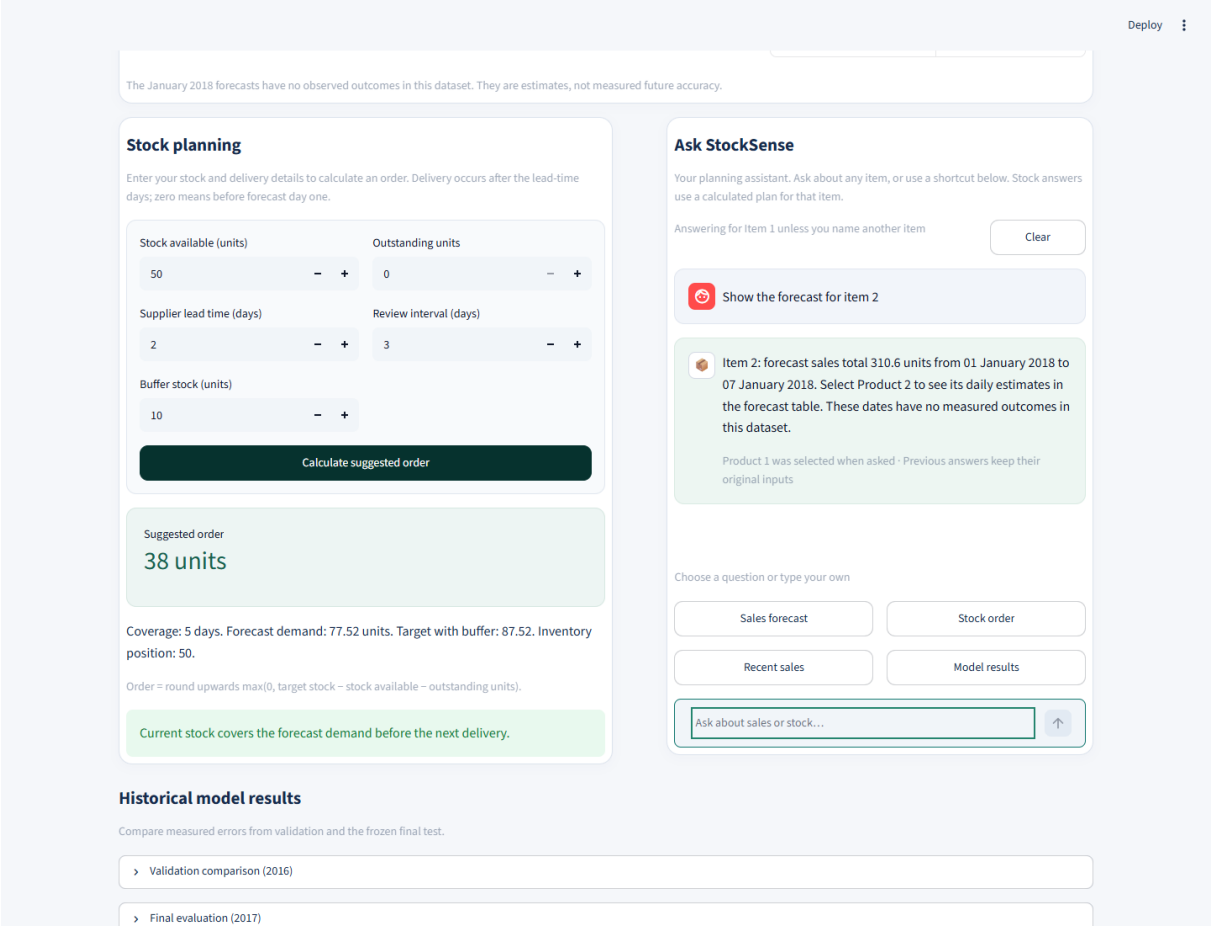


Figure 3. Current office-PC dashboard with demonstration inputs and the resulting 38-unit order suggestion.

The demonstration sequence is item selection, dated forecast, stock calculation, invalid coverage, shortage warning, assistant response and evaluation tables.

12 Conclusion

StockSense AI provides a working five-item sales forecast and stock-planning demonstration through web and terminal interfaces. Random forest won the frozen daily validation criterion and lowered final daily MAE by about 23.3% relative to the weekly benchmark. LSTM's stronger weekly-total result highlights a trade-off relevant to stock decisions. The assistant retrieves the same application objects. Its curated checks passed. Personal-PC rehearsal and the latest 79-test reference run support the technical implementation. Real inventory benefits, general language performance and human usability require further evidence.

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